

Analysis of The Effectiveness of Goodnature Traps on Invasive Species

JAKE NIELSEN
April 2026

New Zealand's unique biodiversity has faced a critical decline since the Polynesian and European settlers introduction of invasive mammal predators. Most notably rats (*Rattus* spp.), stoats (*Mustela erminea*), and brushtail possums (*Trichosurus vulpecula*). These species have disrupted delicate island ecosystems, driving a significant negative trend in native bird, reptile, and insect populations. In response, the New Zealand government launched Predator Free 2050 (PF2050), an ambitious national strategy aimed at the total eradication of these key predators. The success of this initiative is vital for the restoration of New Zealand's natural heritage and the protection of endangered species like the Kiwi and Kākāpō.

A cornerstone of this eradication effort is the transition from labour-intensive traditional methods to smart, ethical trapping technology. Goodnature has engineered the A24 and A12 traps, which represent a positive shift in pest management. Unlike traditional traps that require manual resetting after every strike, these devices are CO₂-powered and self-resetting. A single gas canister allows for up to 24 consecutive strikes without human intervention. This technology is toxin-free and designed to be humane, significantly increasing the efficiency of large-scale trapping networks throughout different terrains.

The motivation for this project lies in the intersection of environmental conservation and data analytics. While the mechanical reliability of the traps is well-established, their ecological impact depends on data-driven deployment. This capstone project, conducted through the Queensland University of Technology (QUT), utilises Goodnature's strike data to investigate the efficacy of these devices in reducing invasive populations. By analysing temporal and spatial trends, this research seeks to optimise trap performance, ensuring that conservation resources are utilised effectively to maximise the protection of native wildlife.

This research utilises a comprehensive dataset provided by Goodnature, consisting of over 1.09 million individual records captured via the A24 trap network between 2018 and 2021. The data, delivered in a CSV format, represents raw telemetry captured by the Goodnature mobile application. While the dataset is "raw," the internal structure is consistent, though it requires pre-processing to address missing values and inconsistencies in certain variables. The dataset is primarily composed of quantitative variables, including strikeTime, temperature, and geospatial coordinates (latitude and longitude). These are complemented by qualitative (categorical) variables such as killType (identifying the species) and activityType. While the volume of data is significant, not all records are complete; therefore, the analysis focuses on high-integrity subsets where key environmental and biological indicators are present.

The weight assigned to specific variables is driven by established biological research. For example, temperature is a known driver of mammal species activity; by correlating temperature with killType, this study aims to model how environmental fluctuations influence pest movement and trap interaction. Furthermore, the geospatial data (latitude and longitude) allows for high-resolution mapping of strike distribution. This spatial analysis is essential for identifying "hotspots" and isolating specific regions for deeper cross-examination with external ecological research.

This study builds upon existing literature regarding the efficacy of self-resetting traps and the historical behaviour of invasive rodents in New Zealand. Previous research into the arrival and proliferation of rats and mice (King 2023) provides a baseline for understanding current population densities, while behavioural studies (Shiels et al., 2022) help validate the "strike rates" observed in the Goodnature data.

The primary objective of this project is to evaluate the effectiveness of Goodnature traps in reducing invasive species populations within the New Zealand ecosystem. This analysis serves as an audit of the technology's real-world impact and provides a data-driven foundation for future product development. To ensure a comprehensive investigation, the following sub-questions guide the research:

1. Environmental Impact: How do temperature variations affect strike rates, and do these patterns reveal seasonal breeding cycles that might outpace eradication efforts?
2. Ecosystem Efficacy: Are the traps more effective in specific terrains or ecosystems, and do those areas align with the preferred habitats of the target species?
3. Population Distribution: Can we establish a clear distribution model for invasive populations based on strike density?
4. Long-term Trends: Since the deployment of these traps, is there a statistically significant reduction in the frequency of strikes, suggesting a declining pest population?

This study transformed raw trap telemetry into a structured dataset to analyse pest population dynamics through a pipeline of cleaning, feature engineering, and statistical modelling.

The primary dataset included trap logs for strikes, mechanical checks, and environmental measurements. To ensure "strike rate" integrity, null values in the killType field were cross-referenced with the activityType column. Entries flagged as 'STRIKE' with null killType were retained as "Unidentified Strikes," while "Test strikes" were purged to prevent human-induced noise from inflating biological data.

Temporal alignment was achieved by converting strikeTime and when columns into standardised datetime objects. Strike Time (CO2 firing) was prioritised; however, when (server receipt) served as a proxy, it was necessary to ensure a continuous record. Raw timestamps were engineered into "Month" and "Season" features, aligning with New Zealand's meteorological calendar: Summer (Dec-Feb), Autumn (Mar-May), Winter (Jun-Aug), and Spring (Sep-Nov).

Spatial engineering was applied using the Haversine formula to assign traps a "Distance from Centroid" and categorise them into ecosystems (Beach, Forest, City, Mountain). An "Out of Range" variable was engineered for data points exceeding a 50km radius to isolate data from rural transition zones. This allowed for analysis of "Invasion Risk" pathways as described by Veale et al. (2011).

Statistical tests were dictated by the discrete, non-normal nature of telemetry data. Pearson Correlation and Linear Regression quantified the relationship between temperature and strike rates. For population distribution, daily counts were modelled as a Probability Density Function (PDF). A Gaussian distribution was rejected because trap strikes are discrete, non-negative "count data." Following Barlow and Barron (2005), a right-skewed model was used to capture the "long tail" of sporadic, high-intensity events.

Finally, an Independent T-test compared daily catch means between 2019 and 2020 (the only full operational periods) to determine if the 18.5% reduction was statistically significant ($p < 0.001$). This p-value allowed for the rejection of the null hypothesis, attributing the decline to successful population suppression rather than random noise, as theorised by King (2023).

Regarding the environmental drivers of pest activity, the analysis demonstrates a moderate positive relationship between temperature variations and A24 trap strike rates, supported by a correlation coefficient of $r = 0.39$. This link is most pronounced during the early phase of the study (2018), as shown in Figure 1, where sharp increases in average monthly temperatures directly preceded significant spikes in trap activity, reaching a peak of nearly 40 strikes per trap. This sharp jump may be linked to data noise or localised measurement anomalies; the recording of this high temperature could not be verified through other sources, like the Earth Science New Zealand data records. Despite this outlier, the broader positive slope in the linear regression model (Figure 2) statistically confirms that as temperatures rise, the probability of a trap strike increases.

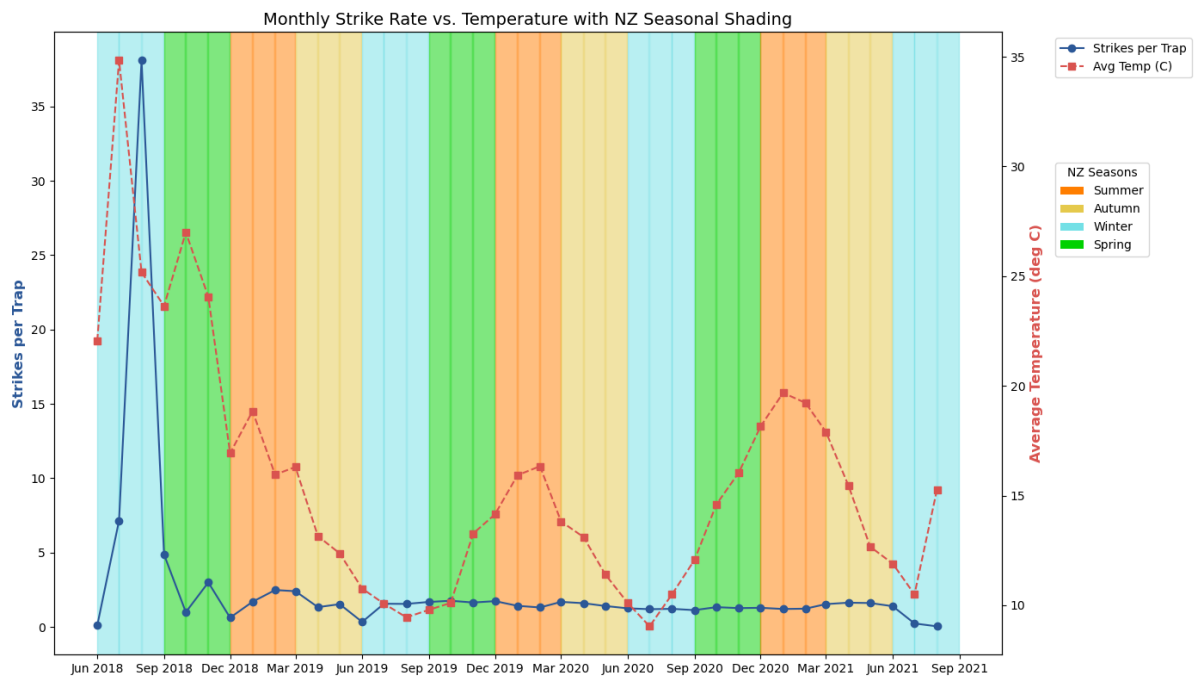


Figure 1: Strike Rate vs. Temperature with NZ Seasons (2018–2021). This time-series visualisation illustrates the relationship between monthly average temperature (red dashed line) and the strike rate per trap (blue solid line). The shaded background represents New Zealand's four seasons.

This correlation likely reflects a combination of biological and environmental drivers. From a "bottom-up" perspective, warmer conditions trigger increases in primary food sources, such as seeds and invertebrates, which subsequently drive the abundance and metabolic activity of the wider predator guild (King, 2023). Furthermore, these seasonal strike patterns may be intensified by the site's high "invasion potential," where proximity to source populations increases the frequency of predator encounters during these active periods (Veale et al., 2011). However, the relationship is not perfectly linear ($R^2 = 0.15$), indicating that while temperature is a significant driver, alternative environmental factors or shifting population densities also play a critical role in determining trap success (Shiels et al., 2022).

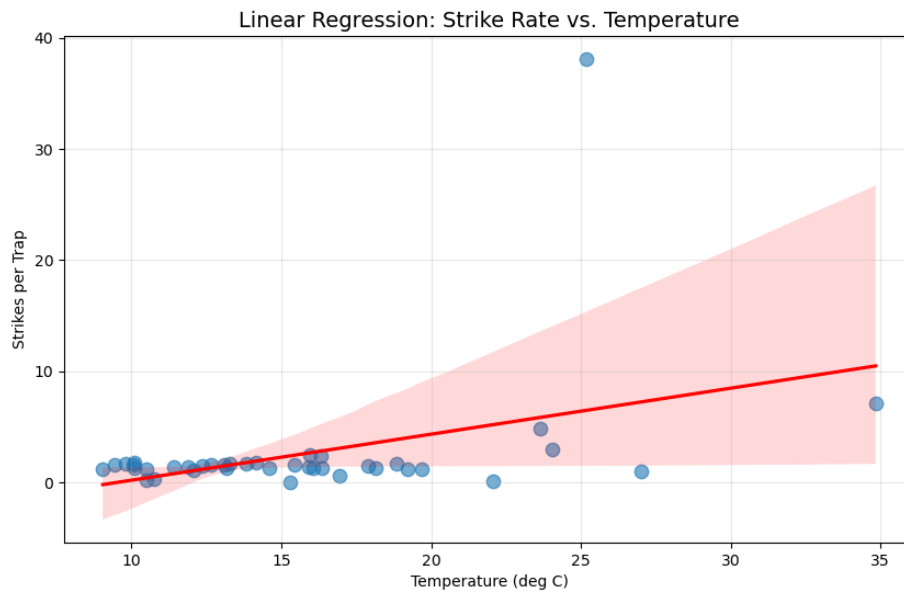


Figure 2: Linear Regression of Temperature vs. Strike Rate. This scatter plot with a regression line ($y = 0.41x - 3.93$) quantifies the relationship between the two variables. The upward slope confirms that higher temperatures generally correlate with higher strike rates, though the spread of data points (residuals) emphasises that temperature is one of several contributing factors.

When considering the spatial distribution of the trap network, the data provides clear insight into how efficacy varies across different ecosystems. The spatial analysis reveals that A24 traps demonstrate the highest absolute effectiveness within City and Beach classifications, which account for most confirmed biological strikes (Figure 3). Urban and coastal environments showed a high concentration of activity, with the city ecosystem specifically recording over 2,500 rat kills. In contrast, the Mountain and Forest classifications yielded significantly lower strike volumes. This disparity suggests that pest density is naturally higher in human-adjacent environments, which often provide superior "habitat quality" and consistent "resource pulsing" compared to high-elevation or dense-canopy zones (King, 2023).

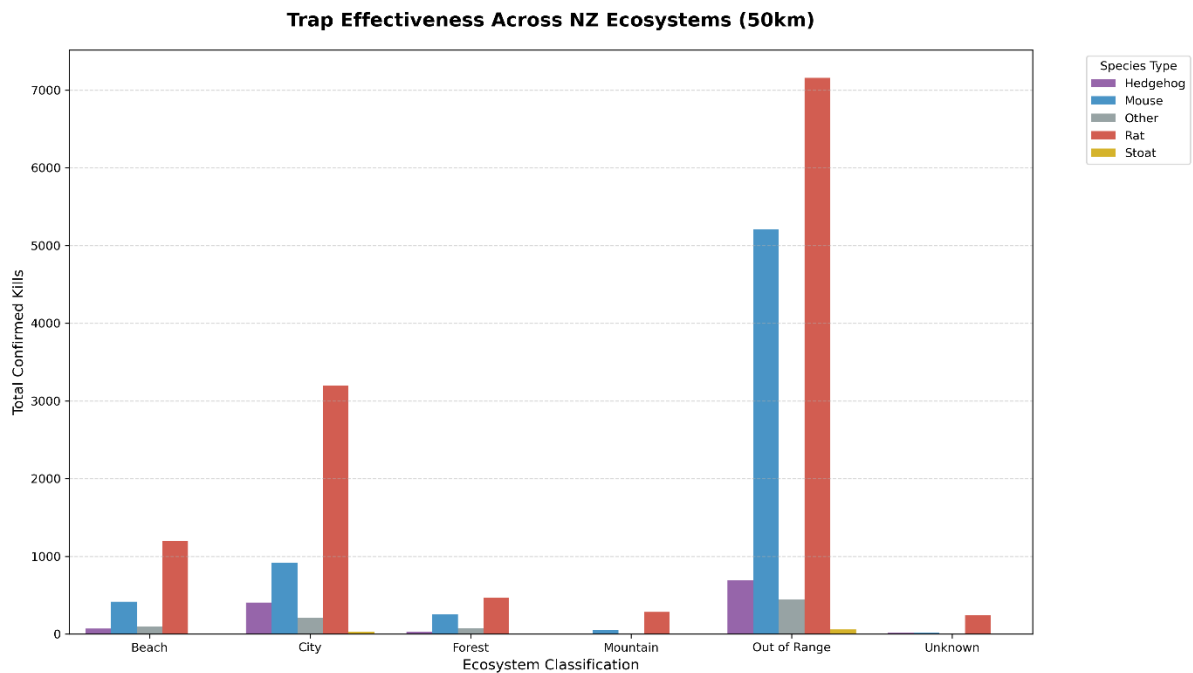


Figure 3: Total Kills by Species and Ecosystem Classification. This grouped bar chart illustrates the breakdown of species strikes across five primary habitat categories: Beach, City, Forest, Mountain, and Out of Range (50km radius cut-off).

The data further shows a strong alignment between ecosystem types and the specific biological positions of the targeted invasive species. Rodents (rats and mice) are the dominant species captured across all classified terrains, but their concentration is nearly five times higher in City and Beach areas than in mountainous regions. This reflects their commensal nature and reliance on human-altered landscapes for food and shelter. Conversely, while smaller in absolute numbers, stoat and hedgehog kills appear more frequently in the Beach and Forest categories. This distribution confirms that the trap network is successfully interacting with the varied biological behaviours of New Zealand's primary pests by intercepting them in the habitats where they are most ecologically active (Shiels et al., 2022).

A primary challenge in this research was the lack of existing literature providing a clear picture of the specific population distributions for these invasive species in the study area. To address this, daily kill counts were modelled as a Probability Density Function (PDF) to establish a quantitative "fingerprint" for the local pest populations (Figure 4). The comparison between the full operational years of 2019 and 2020 demonstrates a visible shift in the distribution; 2019 features a wider variance with a "long tail" extending toward high strike counts, while 2020 shows a more compressed curve centred closer to the origin. This transition suggests that the trapping network successfully shifted the population dynamic from one of frequent surges to a more controlled, low-activity steady state.

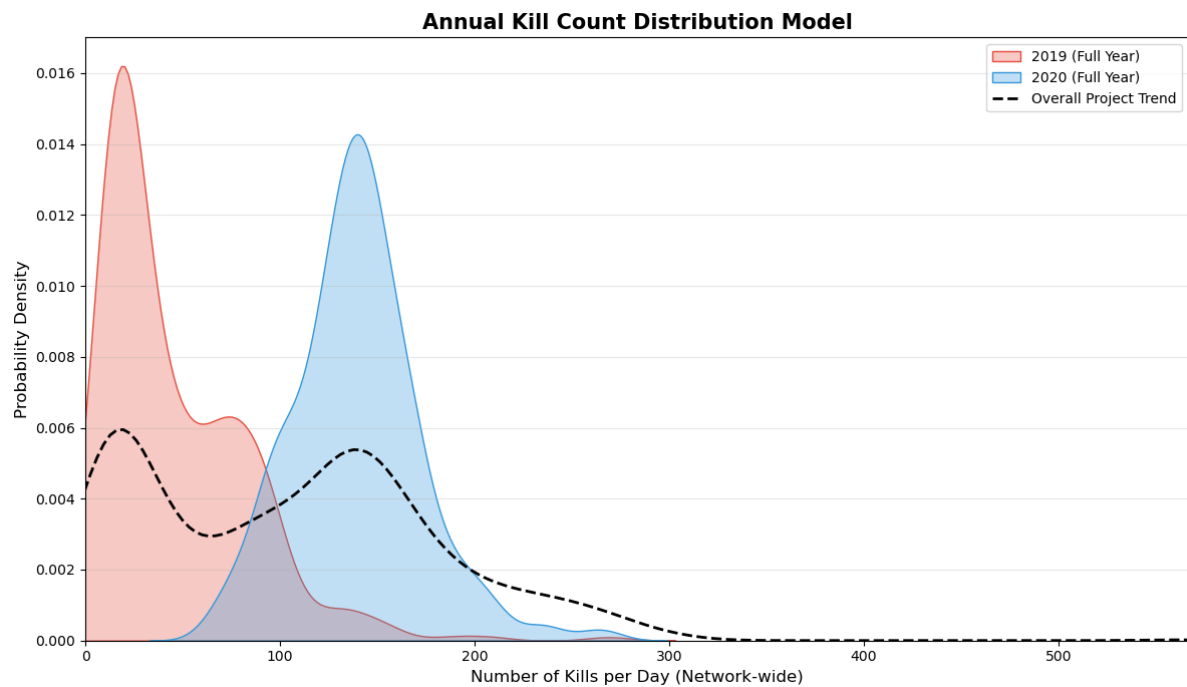


Figure 4: Probability Density Function (PDF) of Daily Kills (2019 vs. 2020). This visualisation compares the kill probability distributions for two consecutive years. The 2019 curve (red) shows a broader spread, indicating a population still capable of high-intensity spikes. The 2020 curve (blue) is taller and narrower. A PDF was chosen over a standard bar chart to visualise the shift in population density as a continuous probability surface.

The Total Project Distribution (2018–2021, black dotted line) exhibits a distinct positive skew, where the median daily kill count is significantly lower than the mean. This "long-tail" nature is consistent with the spatial modelling of Barlow and Barron (2005), which characterises New Zealand’s invasive predators as highly mobile with low-density distributions, leading to intermittent and sporadic trap encounters rather than a uniform stream of activity. Furthermore, variations in the spatial distribution of these kills align with the risk-profiling methods of Veale et al. (2011), suggesting that localised "invasion risks" are heavily influenced by site-specific characteristics. For this study, a successful eradication effort is defined by the ongoing narrowing of this PDF curve, mathematically proving that high-intensity strike days are becoming statistical outliers rather than a predictable part of the seasonal cycle.

The analysis reveals a statistically significant reduction ($p < 0.001$) in the intensity of strikes between the two full operational years of 2019 and 2020. Specifically, the mean daily catch during active periods dropped from 25.26 to 20.59, representing an 18.5% decrease in strike intensity. As shown in Table 1, the p-value is far below the standard alpha threshold ($\alpha = 0.05$), allowing for the rejection of the null hypothesis and confirming that the decline is not due to random seasonal variation.

This 18.5% reduction demonstrates a successful intervention in the "population irruptions" characteristic of New Zealand's invasive mammals (King, 2023). By achieving this level of suppression, the project indicates that the current trapping density is successfully exceeding the intrinsic recruitment and migratory rates modelled by Barlow and Barron (2005). While the A24 is a highly effective tool for such suppression, these results support the findings of Shiels et al. (2022), who noted that the device typically operates as a high-performance management tool rather than a total eradication device.

Metric	2019 Results	2020 Results	Change
Mean Daily catch (Active Days)	25.26	20.59	-18.5%
Standard Deviation	6.55	6.47	---
P-Value	2.13×10^{-9}	----	Highly Significant

Table 1: Statistical Significance of Strike Reduction (2019 vs. 2020). This table summarises the results of the T-test performed on the daily strike data.

The dataset reveals a small but significant relationship ($r \approx 0.39$) between temperature fluctuations and invasive species activity. This correlation provides critical insight for strategic planning, suggesting that maintenance and monitoring efforts can be optimised during winter months when strike rates are at their lowest. Spatial analysis further indicates that high-density areas, such as Cities and Beaches, should remain the primary focus for trap deployment and design. However, the data also highlights untapped potential for eradication within areas between Forests and Mountains. These regions should be audited to ensure that trap spacing is sufficiently dense to intercept highly mobile predators, as suggested by the spatial modelling of Barlow and Barron (2005).

Furthermore, the development of a mathematical distribution model serves as a quantitative "fingerprint" for population health. Goodnature and ecological researchers should continue to refine this model, as it offers a unique lens through which to observe how invasive populations react to new trap designs and increased deployment intensity.

Given the timeframe of the dataset and the observed compression of the strike distribution over time, the effectiveness of the A24 trap is clearly positive. We can confirm that the A24 is not only a functional product but a vital resource in supporting New Zealand's national goal of becoming Predator Free 2050 (PF2050).

Reference list

- **Barlow, N. D., & Barron, M. C. (2005).** Modelling the dynamics and control of stoats in New Zealand forests (Science for Conservation 252). Department of Conservation.

- **King, C. (2023).** Abundance and dynamics of small mammals in New Zealand: Sequential invasions into an island ecosystem like no other. *Life*, 13(1), 156.
<https://doi.org/10.3390/life13010156>

- **Shiels, A. B., Bogardus, T., Crampton, L. H., Gronwald, M., Kreuser, A. M., Baldwin, R. A., & Lepczyk, C. A. (2022).** An introduction to a special issue and review of the effectiveness of Goodnature A24 self-resetting rat traps. *Management of Biological Invasions*, 13(3), 466–478.
<https://doi.org/10.3390/mbi.2022.13.3.01>

- **Veale, A. J., Hannaford, O. D., Russell, J. C., & Clout, M. N. (2011).** Modelling the distribution of stoats on New Zealand offshore islands. *New Zealand Journal of Ecology*, 35(1), 38–48.